

Group 15

Wednesday 17th December, 2025



```
#NOTE:
#there is no 04/2020 for the bike data

import os
os.chdir("..")

def line(char="=", length=50):
    return char * length

import zipfile
import pandas as pd
import glob
import gc

def load_and_aggregate_daily_stats(data_folder="data2"):
    zip_files = sorted(glob.glob(f"{data_folder}/*.zip"))
    print(line())
    print(f"Processing {len(zip_files)} files...")
    print(line())
    daily_sums = {}
    for i, zf in enumerate(zip_files, start=1):
        print(f"[{i}/{len(zip_files)}] {zf}")

        with zipfile.ZipFile(zf, 'r') as z:
            csv_name = z.namelist()[0]
            with z.open(csv_name) as f:
                for chunk in pd.read_csv(f, chunksize=50000, low_memory=False):
                    # Find time column
                    time_col = next((c for c in ['start_time', 'started_at', 'Start Time'] if c in chunk.columns))
                    if not time_col:
                        continue
                    # Find duration column
                    dur_col = next((c for c in ['duration_sec', 'Duration', 'tripduration'] if c in chunk.columns))

                    # Clean dates
                    chunk[time_col] = pd.to_datetime(chunk[time_col], errors='coerce')
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        chunk = chunk.dropna(subset=[time_col])
        chunk['date'] = chunk[time_col].dt.date

    # Aggregate by date
    for date, grp in chunk.groupby('date'):
        if date not in daily_sums:
            daily_sums[date] = {'dur': 0, 'trips': 0, 'bike_share': 0, 'subs': 0}

        if dur_col:
            daily_sums[date]['dur'] += grp[dur_col].sum()
        daily_sums[date]['trips'] += len(grp)
        if 'bike_share_for_all_trip' in grp.columns:
            daily_sums[date]['bike_share'] += (grp['bike_share_for_all_trip'].sum())
        if 'user_type' in grp.columns:
            daily_sums[date]['subs'] += (grp['user_type'] == 'Subscriber').sum()

    if i % 10 == 0:
        gc.collect()

    rows = []
    for date, s in daily_sums.items():
        rows.append({
            'date': date,
            'avg_duration_sec': s['dur'] / s['trips'],
            'pct_bike_share_yes': (s['bike_share'] / s['trips']) * 100,
            'pct_subscriber': (s['subs'] / s['trips']) * 100,
            'total_trips': s['trips']
        })

    df = pd.DataFrame(rows).sort_values('date').set_index('date')
    print(f"\nDone: {len(df)} days, {df['total_trips'].sum():.0f} trips")

    return df

from src.data_utils import load_weather_data
daily_stats = load_weather_data("data/daily_stats.csv")

from src.data_utils import load_weather_data

df = load_weather_data("data/USW00023272.csv")

df["DATE"] = pd.to_datetime(df["DATE"])

mask = (df["DATE"] >= "2018 -01 -01") & (df["DATE"] <= "2024 -12 -31")
df_filtered = df.loc[mask].copy()

print(f"Original rows: {len(df):,}")

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print(f"Filtered rows: {len(df_filtered):,}")  
print(f>Date range: {df_filtered['DATE'].min()} to {df_filtered['DATE'].max()}")
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output_file = "data/weather_2018_2024.csv"  
df_filtered.to_csv(output_file, index=False)  
print(f"Exported to: {output_file}")
```

```
Original rows: 38,326  
Filtered rows: 2,557  
Date range: 2018 -01 -01 00:00:00 to 2024 -12 -31 00:00:00  
Exported to: data/weather_2018_2024.csv
```

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/home/jovyan/final -group15/src/data_utils.py:21: DtypeWarning: Columns (17,19,21,23,25,27,29)  
have mixed dtype. Cast to object.  
return pd.read_csv(path)
```